

Documentation interfaces

This document contains the description of the interfaces used in the file “main_interfaz.f90”.

Note: the variables that contain an asterisk are **optional** arguments.

read_chemistry_interface

- **Description:** Subroutine that reads all the chemical input data (species, reactions, initial and external concentrations, mineral zones, etc.) by dispatching to the appropriate format-specific reader.
- **Class: Chemistry**
- **Input:**

Variable	Type	Description
root	String	Root of input and output files of problem to be solved
path_pb	String	Path to input and output files of problem to be solved
path_DB	String	Path to chemical databases

- **Pre-process:**
 - Sets arbitrary file units for the chemical system, computation options and local chemistry input files.
- **Process:**
 - Selects the input format option (**read_opt**) and reads the **chemical data** accordingly. Currently only the CHEPROO-based format (**read_opt** = 1) is fully implemented; the PHREEQC (**read_opt** = 2) and PFLOTRAN (**read_opt** = 3) branches abort execution with an error stop.

write_conc_comp_wat_types

- **Description:** Subroutine that writes the concentrations of aqueous components of every water type to a file. A reference water type whose name contains **domain**, **initial**, **resident** (or their Spanish equivalents) is located, and its primary-species basis is used to project the variable-activity-species concentrations of every water type onto a common component vector, so all water types yield component vectors of identical dimension.
- **Class: Chemistry**
- **Input:**

Variable	Type	Description
path	String	Path to the output file
filename	String	Output file name

- **Process:**
 - Locates the reference water type by case-insensitive name matching (**domain / initial / resident / dominio / inicial / residente**) and aborts with an error stop if none is found.
 - Opens the output file, writes the names of the reference primary aqueous species, then writes one section per water type containing its component concentrations projected onto the reference basis (in scientific notation, right-aligned).
 - Closes filename

get_num_aq_comps_tar_wat

- **Description:** Function that returns the number of aqueous components in a target water. If no index is provided, the first target water is used.
- **Class: Chemistry**
- **Input:**

Variable	Type	Description
ind_tw*	Integer	Optional index of a target water

- **Output:**

Variable	Type	Description
num_aq_comps	Integer	Number of aqueous components in the target water

- **Process:**
 - Returns the value of **num_aq_prim_species** in the **speciation_alg** object associated to the target water.

interfaz_comps_arch

- **Description:** Subroutine that solves a reactive mixing iteration with **components**, and writes aqueous component and variable activity species concentrations, using **files** as arguments. The integration method is **Euler explicit** and the reaction mixing ratios are **lumped**. It assumes that there are equilibrium reactions.
- **Class: Chemistry**
- **Input:**

Variable	Type	Description
path	String	Path to input and output files.
num_aq_comps	Integer	Nº aqueous components.
file_in	String	File containing concentrations of aqueous components after solving a conservative transport iteration ($\tilde{\mathbf{U}}^{k+1}$).
Delta_t	Real	Time step (Δt^k)
file_out	String	File where aqueous component concentrations ($\mathbf{U}^{k+1}$) and variable activity species concentrations ($\mathbf{C}_v^{k+1}$) after solving a reactive mixing iteration will be written.

- **Pre-process:**
 - Allocate aqueous component concentrations after conservative transport ($\tilde{\mathbf{U}}^{k+1}$)
 - Allocate chemical reactions contribution to aqueous component concentrations ($\mathbf{U}_K^{k+1}$)
- **Process:**
 - **Read $\tilde{\mathbf{U}}^{k+1}$ from file_in. Note:** The rows must represent components and the columns targets.
 - **Solve** reactive mixing iteration for each target water j :
 - **Loop over number of target waters**
 - Compute chemical reactions contribution to aqueous component concentrations using **Euler explicit** and **lumping**:
$$\mathbf{u}_{K,j}^{k+1} := \Delta t^k \mathbf{W} \mathbf{S}_{K,v}^T \mathbf{r}_{K,j}^k$$
 - Update the total aqueous component concentrations in target water: $\mathbf{u}_j^{k+1} = \tilde{\mathbf{u}}_j^{k+1} + \mathbf{u}_{K,j}^{k+1}$
 - Speciate to compute variable activity species concentrations $\mathbf{c}_{v,j}^{k+1}$ from components, using Newton-Raphson.
 - Check Newton convergence: if speciation does not converge for a target water, the target index and a "No convergence in speciation after reactive mixing iteration. Try reducing the time step" message are printed and execution is aborted with an error stop.

- **Post-process:**

- **Write $\mathbf{U}^{k+1}$ in file_out** (same dimensions as $\tilde{\mathbf{U}}^{k+1}$)
- **Write $\mathbf{C}_v^{k+1}$ in file_out** (rows: species, columns: targets)
- **Deallocate arrays**

interfaz_esp_arch

- **Description:** Subroutine that solves a reactive mixing iteration with **aqueous species**, and writes aqueous species concentrations, using **files** as arguments. The integration method is **Euler explicit** and the reaction mixing ratios are **lumped**. It assumes that there are NO equilibrium reactions.

- **Class: Chemistry**

- **Input:**

Variable	Type	Description
path	String	Path to input and output files.
num_aq_comps	Integer	Nº aqueous species (=nº aqueous components).
file_in	String	File containing concentrations of aqueous species after solving a conservative transport iteration ($\tilde{\mathbf{C}}^{k+1}$).
Delta_t	Real	Time step (Δt^k)
file_out	String	File where aqueous species concentrations ($\mathbf{C}^{k+1}$) after solving a reactive mixing iteration will be written.

- **Pre-process:**

- Allocate aqueous species concentrations after conservative transport ($\tilde{\mathbf{C}}^{k+1}$)
- Allocate chemical reactions contribution to aqueous species concentrations ($\mathbf{C}_K^{k+1}$)

- **Process:**

- **Read** $\tilde{\mathbf{C}}^{k+1}$ from file_in. **Note:** The rows must represent species and the columns targets.
- **Solve** reactive mixing iteration for each target water j :

- **Loop over number of target waters**

- Compute chemical reactions contribution to aqueous species concentrations using **Euler explicit** and **lumping**:

$$\mathbf{c}_{K,j}^{k+1} := \Delta t^k \mathbf{S}_{K,aq}^T \mathbf{r}_{K,j}^k$$

- Update the total aqueous species concentrations in target water:

$$\mathbf{c}_j^{k+1} = \tilde{\mathbf{c}}_j^{k+1} + \mathbf{c}_{K,j}^{k+1}$$

- **Post-process:**

- **Write** $\mathbf{C}^{k+1}$ in file_out (rows: species, columns: targets)
- **Deallocate** arrays